Blog Notes 3

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Introduction

- Stochastic Differential Equation
- Stochastic Calculus V.S. Real Calculus
- Itô Formula
- Brownian Motion
- Geometric Brownian Motion
- Hull-White Interest Rate model

Notes:

In this note, I would write my understanding of Solving stochastic integral, using Itô Formula and how to apply it to solve three SDE examples(Brownian Motion, Geometric Brownian Motion, one Interest Rate model).

Stochastic Differential Equation

A generic form of a stochastic differential equation is given by

$$dX(t) = \beta(t, X(t))dt + \gamma(t, X(t))dW(t)$$

where,

- $\beta(t, X(t))$ is called a drift,
- $\gamma(t, X(t))$ is called a diffusion.
- $W(t)$ is a Brownian motion.

Stochastic Differential Equation

Examples

- Dynamics of a Brownian Motion is $dX(t) = \mu dt + \sigma dW(t)$, where $\beta(t, X(t)) = \mu, \gamma(t, X(t)) = \sigma$.

- Dynamics of a Geometric Brownian Motion(GBM) is $dX(t) = \mu X(t)dt + \sigma X(t)dW(t)$. where $\beta(t, X(t)) = \mu X(t), \gamma(t, X(t)) = \sigma X(t)$.
- The Dynamics of a GBM can be written as $\frac{dX(t)}{dt} = \mu dt + \sigma dW(t)$, and can be observed that it can model the Dynamics of Returns on an Asset Price $X(t)$.

- Dynamics of a Hull-White Interest model is $dR(t) = (a(t) - b(t) \cdot R(t))dt + \sigma(t) \cdot d\widetilde{W}(t)$ where $\beta(t, R(t)) = a(t) - b(t) \cdot R(t), \gamma(t, R(t)) = \sigma(t)$.

Stochastic Differential Equation

When given a stochastic differential equation which models dynamics of a stochastic process, in generic form as follows,

$$dX(t) = \beta(t, X(t))dt + \gamma(t, X(t))dW(t)$$

Together with an given initial condition as

$$X(t) = x, \quad t \geq 0 \quad \text{and} \quad x \in \mathbb{R}$$

Question: What is solving the stochastic differential equation?

Answer: Solving the SDE is to solve for a function $X(T)$, which is a stochastic process that satisfies the SDE and the initial condition.

Idea: Integrate both sides, from time t to time T;

$$\int_t^T dX(s) = \int_t^T \beta(s, X(s))ds + \int_t^T \gamma(s, X(s))dW(s)$$

Stochastic Calculus V.S. Real Calculus

Solving stochastic differential equation involves integrals with respect to stochastic process and integrals with respect to time, so it involves stochastic calculus and real calculus.

Example:

take the Itô integral

$$\int_t^T dW(s), \text{ where, } W(t) \text{ is a standard Brownian motion}$$

as an example to see the difference between stochastic calculus and real calculus.

Stochastic Calculus V.S. Real Calculus

Two ways to explain what the stochastic integral $\int_t^T dW(s)$.

1. $\int_t^T dW(s)$ means adding up all the little changes $dW(s)$ from time t to time T.

   Assume a small time interval Δ, sum up all the changes of $W(s)$ from t to T in the small consecutive intervals and then make the time interval smaller and smaller. That is,

$$
\begin{aligned}
& \int_t^T dW(s) \\
= \quad & \lim_{\Delta \to 0} [(W_{t+\Delta} - W_t) + (W_{t+2\Delta} - W_{t+\Delta}) + \ldots + (W_T - W_{T-\Delta})] \\
= \quad & \lim_{\Delta \to 0} (W_T - W_t) = W(T) - W(t)
\end{aligned}
$$

Stochastic Calculus V.S. Real Calculus Continue

Corresponding to the stochastic integral $\int_t^T dW(s)$, in real calculus, an integral interested is $\int_t^T \frac{dW(s)}{ds} ds$.

In real calculus, the integral means calculating the area under the curve chopped in the time interval dt.

Because Brownian motion $W(t), t \geq 0$ has rough(non-smooth) trajectories with respect to time and it fluctuates a lot in short period, there is no smooth function under which the area can be taken.

Stochastic Calculus V.S. Real Calculus Continue

To calculate the integral of the non-smooth (non-differential) curve $W(t)$, use the notation $dW(t)$.

In real calculus, the conventional notation for the integral with respect to time is typically for smooth function.

However, **in stochastic calculus**, the idea is adding up all the little changes of the process, $dW(t)$, and get the level of the change of the process $W(t)$ from time t to time T, that is,

$$\int_t^T dW(s) = W(T) - W(t)$$

Stochastic Calculus V.S. Real Calculus Continue

There is another way to explain what the stochastic integral $\int_t^T dW(s)$.

2. $\int_t^T dW(s) = W(T) - W(t)$ is a random variable.

   By the definition of Brownian motion, its increment $W(T) - W(t)$ is a normally distributed random variable with

$$\begin{cases} \text{Mean} & = & E(W_T - W_t) = 0 \\ \text{Variance} & = & Var(W_T - W_t) = T - t \end{cases}$$

   That is,

$$W_T - W_t = N(0, T - t) = \sqrt{(T - t)}N(0, 1)$$

   Actually,

$$dW(t) = \sqrt{dt}, \quad (dW(t))^2 = dt$$

Stochastic Calculus

Principal to Keep in Mind:

The basic principal to keep in mind is that **stochastic calculus** is like a regular calculus, but it do second order Taylor expansions and keep the term dt, $dW(t)$ and $(dW(t))^2$ ($(dW(t))^2$ is the same as dt term, because $(dW(t))^2 = dt$), omit the higher order terms.

In contrast, **real calculus** do the first order Taylor expansions, only keep dt term and don't keep higher order terms.

Basic Lemma to use in Itô formula:

In the Itô formula, take the differential to second order expansions and omit the terms $dtdW(t)$, $dtdt$, simplify the term $dW(t)dW(t)$, by letting

$$\begin{cases} dtdW(t) &= 0, \quad dtdt = (dt)^2 = 0 \\ dW(t)dW(t) &= (dW(t))^2 = dt \end{cases}$$

Itô Formula

An important multivariate version Itô Formula

In stochastic calculus, suppose $\{X_t, t \geq 0\}$, $\{Y_t, t \geq 0\}$ are two Itô process, for a C^2-function $f(x, y)$ which is twice continuously differentiable function, then, the composition of $f(x, y)$ applied to $X(t)$ and $Y(t)$ is again an Itô process, with the differential given by

$$\begin{aligned} & df(X_t, Y_t) \\ = & \frac{\partial f}{\partial X} dX + \frac{\partial f}{\partial Y} dY + \frac{\partial^2 f}{\partial X^2} dXdX + \frac{\partial^2 f}{\partial Y^2} dYdY + \frac{\partial^2 f}{\partial X \partial Y} dXdY \\ = & f_X dX + f_X dX + \frac{1}{2} f_{X,X} (dX)^2 + \frac{1}{2} f_{Y,Y} (dY)^2 + f_{X,Y} dXdY \end{aligned}$$

In the above formula, use simplified notation dX, dY representing $dX(t)$, $dY(t)$. It is noted that the term $\frac{\partial^2 f}{\partial X \partial Y} dXdY$ is from $\frac{1}{2} \frac{\partial^2 f}{\partial X \partial Y} dXdY + \frac{1}{2} \frac{\partial^2 f}{\partial Y \partial X} dYdX$, where $\frac{\partial^2 f}{\partial X \partial Y} dXdY = \frac{\partial^2 f}{\partial Y \partial X} dYdX$.

Itô Formula Continue

In the two variable version of Itô Formula, $\{X_t, t \geq 0\}$, $\{Y_t, t \geq 0\}$ are two Itô processes, in the differential of $df(X(t), Y(t))$, there are terms $dX(t)$, $dY(t)$, $dX(t)dX(t)$, $dY(t)dY(t)$, $dX(t)dY(t)$.

Following the basic lemma to use in the Itô formula, that is,

$$\begin{cases} dtdW(t) &= 0, \quad dtdt = (dt)^2 = 0 \\ dW(t)dW(t) &= (dW(t))^2 = dt \end{cases}$$

In the multivariate version Itô Formula, if
$dX(t) = \mu_1(t)dt + \sigma_1(t)dW(t)$ and
$dY(t) = \mu_2(t)dt + \sigma_2(t)dW(t)$, where both are decomposed to dt term and $dW(t)$ term,
then, **another lemma to simplify the formula** is

$$\begin{cases} dX(t)dX(t) &= (\sigma_1(t)dW(t))^2 = \sigma_1^2(t)dt, \\ dY(t)dY(t) &= (\sigma_2(t)dW(t))^2 = \sigma_2^2(t)dt, \\ dX(t)dY(t) &= \sigma_1(t)\sigma_2(t)(dW(t))^2 = \sigma_1(t)\sigma_2(t)dt \end{cases}$$

Common Formula Example in Stochastic Calculus:

In stochastic calculus, we have

$$df(X,Y) = f_X dX + f_Y dY + \frac{1}{2} f_{X,X}(dX)^2 + \frac{1}{2} f_{Y,Y}(dY)^2 + f_{X,Y} dX dY$$

Example1:

Suppose $f(X,Y) = X \cdot Y$, we have

$$d(XY) = YdX + XdY + dXdY$$

which follows the stochastic calculus formula $df(X,Y)$ and from

$$\begin{cases} f(X,Y) = XY, \\ f_X = Y, \quad f_Y = X, \\ f_{X,X} = 0, \quad f_{Y,Y} = 0, \quad f_{X,Y} = 1, \quad f_{Y,X} = 1 \end{cases}$$

Common Multivariate Itô Formula Examples:

In the Itô formula, the aim is to find a stochastic differential of a composition of C^2-function applied to Itô process,

Example2: two variable version Itô formula

Suppose the composition of a twice continuously differentiable function $f(x, y) = xy$ is applied to two Itô processes

$$dX_i(t) = \mu_i(t)dt + \sigma_i(t)dW(t), \quad i = 1, 2$$

The product $X_1(t)X_2(t)$ is again an Itô process.
Find: $d(X_1(t)X_2(t))$.
Answer:

$$d(X_1(t)X_2(t)) = X_1(t)dX_2(t) + X_2(t)dX_1(t) + \sigma_1(t)\sigma_2(t)dt$$

which follows the stochastic calculus formula $d(XY)$ and from

$$\begin{cases} d(XY) &= YdX + XdY + dXdY, \\ dX_1(t)dX_2(t) &= \sigma_1(t)\sigma_2(t)(dW(t))^2 = \sigma_1(t)\sigma_2(t)dt \end{cases}$$

Common Multivariate Itô Formula Examples:

Example 3: two variable version Itô formula, one variable is time t

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$ and suppose an C^2-function $f(x, y)$.

Find: $d(f(t, X(t)))$.

Answer:

$$\begin{aligned} d(f(t, X(t))) &= f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\ &= f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \end{aligned}$$

which follows the stochastic calculus formula $df(X, Y)$ and from

$$\begin{cases} (dt)^2 = 0, \quad dtdX(t) = 0 \\ \text{[The term } dtdX(t) \text{ contains } (dt)^2 = 0, dtdW(t) = 0] \\ dX(t)dX(t) = (\sigma(t)dW(t))^2 = \sigma^2(t)dt \end{cases}$$

Single variable version Itô formula

An important one variable version Itô formula

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$ and suppose an C^2-function $f(x)$. Then, the composition of $f(x)$ applied to the Itô process $\{X(t), t \geq 0\}$ is again an Itô process with the differential given by

$$\begin{aligned} d(f(X(t)) &= f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\ &= f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \end{aligned}$$

which follows the stochastic calculus formula $df(X, Y)$ and from

$$\begin{cases} f_t = 0, \quad f_{t,t} = 0 \quad f_{t,x} = f_{x,t} = 0 \\ \text{[No need to consider terms } dtdt, \ \ dtdX(t) or dX(t)dt] \\ dX(t)dX(t) = (\sigma(t)dW(t))^2 = \sigma^2(t)dt \end{cases}$$

Compared to Real Calculus

Note1: In real calculus, suppose a function $f(x)$ and a function $X(t)$, the composition of $f(x)$ applied to the function $X(t)$, the differential of $f(X(t))$ is $df(X(t)) = f_X dX(t)$ and this would only hold if $X(t)$ has differentiable trajectories.

2: In the one variable version Itô formula, recall that the term $\sigma(t)dW(t)$ relates to the stochastic integral $\int_t^T \sigma(t)dW(s)$, which has rough trajectories (not differentiable trajectories) and inherited by $X(t)$, and it causes the second order correction term. It also explains why this one variable version Itô formula only works for C^2-functions $f(x)$ (twice continuously differentialble function).

Compared to Real Calculus

Note3: Similarly, in real calculus, for multivariate function $f(x, y) = xy$ applied to two deterministic functions $X_1(t)$, $X_2(t)$. The differential is the product rule $dX_1(t)X_2(t) = X_1(t)dX_2(t) + X_2(t)dX_1(t)$, which is from the diffential of $f(x, y)$ in real calculus; $df(x, y) = f_y dy + f_x dx$.

It only holds for $X_1(t)$, $X_2(t)$ had no rough trajectories (differentiable trajectories). But due to the stochastic integral $\int_t^T \sigma_i(t)dW(s)$ inherited by $X_i(t)$, then we have the second order correction term.

Common single variable Itô Formula Examples:

Example 1: Exponential $f(x) = e^x$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An exponential function $f(x) = e^x$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by $Y(t) = e^{X(t)}$.

Find: $dY(t)$.

Answer: Because $Y(t)$ is a smooth function (differentiable to any higher order, including order 2), we can use one variable Itô formula, the decomposition of $dY(t)$ is given by

Common single variable Itô Formula Examples:

$$
\begin{aligned}
d(Y(t)) &= d(e^{X(t)}) = f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \\
&= e^{X(t)} dX(t) + \frac{1}{2} e^{X(t)} \sigma^2(t) dt \\
&= Y(t) dX(t) + \frac{1}{2} Y(t) \sigma^2(t) dt
\end{aligned}
$$

which follows the stochastic calculus formula $df(X(t))$ and from

$$
\begin{cases}
f(x) = e^x, \quad f_x = e^x \;\; f_{x,x} = e^x \\
\\
df(X(t)) = f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\
\qquad\qquad = f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt
\end{cases}
$$

Common single variable Itô Formula Examples:

Definition

Motivated from the example, Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.

there is a counter-part to the classical exponential function in stochastic calculus, which would have a same formula as the differential of exponential in real calculus. It is called the stochastic exponential.

Denoted by $\mathcal{E}(X(t))$, **the stochastic exponential** is defined by

$$\mathcal{E}(X(t)) = e^{X(t) - \frac{1}{2}\int_0^t \sigma^2(s)ds}$$

Fact:

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$. $\mathcal{E}(X(t))$ satisfy the stochastic differential equation(SDE): $d\mathcal{E}(X(t)) = \mathcal{E}(X(t))dX(t)$

Common single variable Itô Formula Examples:

Motivation idea

The way to define the stochastic exponential is that we would like have the form $d(Y(t)) = d(e^{X(t)}) = e^{X(t)}dX(t)$ same as $d(e^{x(t)}) = e^{x(t)}dx(t)$, not appear the second order correction term $\frac{1}{2}Y(t)\sigma^2(t)dt$.

The way is to assume an exponential $Y(t)$ where after applying Itô formula, it has the decomposition

$$dY(t) = -\frac{1}{2}Y(t)\sigma^2(t)dt + Y(t)dX(t) + \frac{1}{2}Y(t)\sigma^2(t)dt$$

Common single variable Itô Formula Examples:

Motivation idea Continue

We know that if the exponential is $Y(t) = e^{X(t)}$, its decomposition is $Y(t)dX(t) + \frac{1}{2}Y(t)\sigma^2(t)dt$, our aim is to construct another form of exponential $Y(t)$ such that it has one more term $-\frac{1}{2}Y(t)\sigma^2(t)dt$ after using Itô formula.

The decomposition of the exponential differential has three terms, consider the Itô formula
$d(f(t, X(t))) = f_t dt + f_X dX(t) + f_{X,X}\sigma^2(t)dt$

We would like $f(t, X(t))$ be an exponential, compared to the formula $d(f(X(t))) = d(e^{X(t)}) = e^{X(t)}dX(t) + \frac{1}{2}e^{X(t)}\sigma^2 dt$

Then, we would be motivated to have the exponential defined by $f(t, X(t)) = e^{X(t) - \frac{1}{2}\int_0^t \sigma^2(s)ds}$, denoted the exponential by $\mathcal{E}(X(t))$, and call it **the stochastic exponential**.

Common single variable Itô Formula Examples:

Example 2: Check the Itô formula of the Exponential $f(t,x) = e^{x-\frac{1}{2}\int_0^t \sigma^2(s)ds}$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An exponential function $f(t,x) = e^{x-\frac{1}{2}\int_0^t \sigma^2(s)ds}$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by
$Y(t) = f(t, X(t)) = e^{X(t)-\frac{1}{2}\int_0^t \sigma^2(s)ds}$.

Find: $dY(t)$.

Answer:
$f_t(t,X) = -\frac{1}{2}\sigma^2(t)Y(t)$, $f_X(t,X) = Y(t)$, $f_{X,X}(t,X) = Y(t)$

Apply the Itô formula $df(t, X(t)) =$
$f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X}(dX(t))^2 = f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X}\sigma^2(t)dt$,

We get: $dY(t) = Y(t)dX(t)$

Common single variable Itô Formula Examples:

Example 2: Logarithmic function $f(x) = log(x)$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An logarithmic function $f(x) = log(x)$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by $f(X(t)) = logX(t)$.

Find: $d\ logX(t)$.

Answer:
$f_X(X) = \frac{1}{X(t)}$, $f_{X,X}(X) = \frac{-1}{(X(t))^2}$

Apply the Itô formula
$df(X(t)) = f_X dX(t) + \frac{1}{2} f_{X,X}(dX(t))^2 = \frac{dX(t)}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2$,

We get: $d\ logX(t) = \frac{dX(t)}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2 = \frac{dX_t}{X_t} - \frac{1}{2}\left(\frac{dX_t}{X_t}\right)^2$

Back to Stochastic Differential Equation

When given a stochastic differential equation which models dynamics of a stochastic process, in generic form as follows,

$$dX(t) = \beta(t, X(t))dt + \gamma(t, X(t))dW(t)$$

Together with an given initial condition as

$$X(t) = x, \quad t \geq 0 \quad \text{and} \quad x \in \mathbb{R}$$

Question: Solving the stochastic differential equation

Task: Solve for the solution $X(T)$. which is a stochastic process that satisfies the SDE and the initial condition.

Idea: Integrate both sides, from time t to time T;

$\int_t^T dX(s) = \int_t^T \beta(s, X(s))ds + \int_t^T \gamma(s, X(s))dW(s)$

Back to Stochastic Differential Equation

$\leftrightarrow$ $X(T) - X(t) = \int_t^T \beta(s, X(s))ds + \int_t^T \gamma(s, X(s))dW(s)$

From the initial condition $X(t) = x$,

We get the solution of the stochastic differential equation;
$X(T) = x + \int_t^T \beta(u, X(u))du + \int_t^T \gamma(u, X(u))dW(u)$

Note 1: $X(T) =$ initial condition $+$ ordinary integral $+$ Itô integral

2: Given the value of $X(t)$, from the decomposition of $X(T)$, $X(T)$ is independent of the filtration $\mathcal{F}_t$ and depends on the possible paths of Brownian motion between t and T.

Back to Stochastic Differential Equation

Note3: Suppose a function $h(x)$ applied to the stochastic process $X(T)$. Consider the conditional expectation of $h(X(T))$ at time t; $E^{t,x}[h(X(T))] = E_t[h(X(T))|X(t) = x]$
Then, it depends on the paths of Brownian motion between t and T. It is a function of t and x, can be denoted by $g(t, x)$.

4: Consider the expectation of $h(X(T))$ at time 0, with the given information $\mathcal{F}(t)$; $E_0[h(X(T))|\mathcal{F}(t)] = E[h(X(T))|\mathcal{F}(t)]$, the value of $X(t)$ is not known at time $t = 0$, the expectation depends on the value of x and t, can be denoted by $g(t, X(t))$.

5: Actually, without a given initial condition, the stochastic process $X(T)$ is a **Markov**, because it only depends on the present value of $X(t)$ and are independent of the previous values of the process $\{X(s), s < t\}$.

Stochastic Differential Equation Examples

Example 1:

Dynamics of a Brownian Motion is $dX(t) = \mu dt + \sigma dW(t)$, where $\beta(t, X(t)) = \mu, \gamma(t, X(t)) = \sigma$. Solve the stochastic differential equation.

Answer: integrate both sides, from time t to time T;

$\int_t^T dX(s) = \int_t^T \mu ds + \int_t^T \sigma dW(s) = \mu \int_t^T ds + \sigma \int_t^T dW(s)$

$\leftrightarrow$

$$\begin{cases} X(T) - X(t) & = & \mu(T-t) + \sigma \int_t^T dW(s) \\ & = & \mu(T-t) + \sigma (W(T) - W(t)) \\ & = & \mu(T-t) + \sigma N(0, T-t) \\ & = & N(\mu(T-t), \sigma^2(T-t)) \end{cases}$$

From the equation, the change of values of the stochastic process $\{X(t), t \geq 0\}$ is normally distributed with mean $\mu(T-t)$ and variance $\sigma^2(T-t)$.

Stochastic Differential Equation Examples

Example 2:

Dynamics of returns on an asset price $X(t)$ is modelled by a Geometric Brownian Motion(GBM)
$dX(t) = \mu X(t)dt + \sigma X(t)dW(t)$. where the drift term
$\beta(t, X(t)) = \mu X(t)$, the noise(martingale) term
$\gamma(t, X(t)) = \sigma X(t)$. Solve the stochastic differential equation.

Answer:
Change the dynamic equation to $\frac{dX(t)}{X(t)} = \mu dt + \sigma dW(t)$,

Consider $d\ logX(t)$ using Itô lemma. This is because the decomposition of $d\ logX(t)$ would contain a term $\frac{dX(t)}{X(t)}$ and a second order correction term in stochastic calculus.

From the single variable Itô formula example 2 of logarithmic function $f(x) = log(x)$ applied to an Itô process, we have
$d\ logX(t) = \frac{dX(t)}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2 = \frac{dX_t}{X_t} - \frac{1}{2}\left(\frac{dX_t}{X_t}\right)^2$

Stochastic Differential Equation Examples

Continue Example 2:

Substitute the dynamic equation $\frac{dX(t)}{X(t)} = \mu dt + \sigma dW(t)$ into $d\ logX(t) = \frac{dX(t)}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2$,

we get: $\quad d\ logX(t) = \mu dt + \sigma dW(t) - \frac{1}{2}\left(\mu dt + \sigma dW(t)\right)^2$

From Itô formula lemma, we have $(dt)^2 = 0$, $dtdW(t) = 0$, $(dW(t))^2 = dt$, use them to simplify $(\mu dt + \sigma dW(t))^2$, get

$$\begin{aligned}(\mu dt + \sigma dW(t))^2 &= \mu^2(dt)^2 + \sigma^2\left(dW(t)\right)^2 + 2\mu\sigma dtdW(t) \\ &= \sigma^2 dt\end{aligned}$$

then, we get:

$$\begin{aligned}d\ logX(t) &= \mu dt + \sigma dW(t) - \tfrac{1}{2}\sigma^2 dt \\ &= \left(\mu - \tfrac{1}{2}\sigma^2\right) dt + \sigma dW(t)\end{aligned}$$

Stochastic Differential Equation Examples

Continue Example 2:

integrate both sides, from time t to time T;

$$\begin{aligned} \int_t^T d\ logX(t) &= \int_t^T \left(\mu - \tfrac{1}{2}\sigma^2\right) ds + \int_t^T \sigma dW(s) \\ &= \left(\mu - \tfrac{1}{2}\sigma^2\right) \int_t^T ds + \sigma \int_t^T dW(s) \end{aligned}$$

$\leftrightarrow$

$$\begin{cases} logX(T) - logX(t) &= \left(\mu - \frac{1}{2}\sigma^2\right)(T-t) + \sigma \int_t^T dW(s) \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\left(W(T) - W(t)\right) \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)(T-t) + \sigma N(0, T-t) \\ &= N\left(\left(\mu - \frac{1}{2}\sigma^2\right)(T-t),\ \sigma^2(T-t)\right) \end{cases}$$

From the equation, the change of log values of the stochastic process $\{X(t), t \geq 0\}$ is normally distributed with mean $\left(\mu - \frac{1}{2}\sigma^2\right)(T-t)$ and variance $\sigma^2(T-t)$.

the asset price is $log(\frac{X(T)}{X(t)}) = \left(\mu - \frac{1}{2}\sigma^2\right)(T-t) + \sigma \int_t^T dW(s)$

In the end, we have
$X_T = X_t e^{\left(\mu - \frac{1}{2}\sigma^2\right)(T-t) + \sigma \int_t^T dW(s)} = X_t e^{N\left(\left(\mu - \frac{1}{2}\sigma^2\right)(T-t),\ \sigma^2(T-t)\right)}$

Stochastic Differential Equation Examples

Example 3:

In Hull-White model, the dynamics of interest model is given by
$dR(t) = (a(t) - b(t) \cdot R(t))dt + \sigma(t) \cdot d\widetilde{W}(t)$
where $a(t)$, $b(t)$ and $\sigma(t)$ are basically the non-random positive functions. $\widetilde{W}(t)$ is a Brownian motion under the risk-neutral probability measure. Given an initial condition $R(t) = r$.

Find the solution of the stochastic differential equation.

Answer:

Idea: Look at the differential $d(e^{\int_0^t b(s)ds} \cdot R(t))$ using one common multivariate Itô Formula example
$d(X(t)Y(t)) = Y(t)dX(t) + X(t)dY(t) + dX(t)dY(t)$.
(details proof of the formula can be seen in Example 1 and Example 2 of common multivariate Itô Formula examples.)

Stochastic Differential Equation Examples

Denote $e^{\int_0^t b(s)ds}$ by $X(t)$, compute $dX(t)$.

$b(t)$ is a non-random function, use ordinary calculus, we have
$dX(t) = e^{\int_0^t b(s)ds} b(t)dt = X(t)b(t)dt$.

Next, compute the differential $d(e^{\int_0^t b(s)ds} \cdot R(t)) = d(X(t) \cdot R(t))$
using the common multivariate multivariate Itô Formula example
$d(X(t)Y(t))$, we have
$d(X(t) \cdot R(t)) = X(t)dR(t) + R(t)dX(t) + dX(t)dR(t)$,

Using the Itô multiplication table, compute $dX(t)dR(t)$, it equals
$X(t)b(t)dt \cdot \Big((a(t) - b(t) \cdot R(t))dt + \sigma(t) \cdot d\widetilde{W}(t)\Big) =$
$(dt)^2$ term $+ dtd\widetilde{W}(t)$ term $= 0$, then, we have
$d(X(t) \cdot R(t)) = X(t)dR(t) + R(t)dX(t)$

Stochastic Differential Equation Examples

Next, substitute $dR(t) = (a(t) - b(t) \cdot R(t))dt + \sigma(t) \cdot d\widetilde{W}(t)$ into $d(X(t) \cdot R(t))$, we have

$$\begin{aligned} & d(X(t)R(t)) \\ = \; & X(t)[a(t) - b(t) \cdot R(t)]dt + X(t)\sigma(t)d\widetilde{W}(t) + R(t)X(t)b(t)dt \\ = \; & X(t)[a(t)dt + \sigma(t)d\widetilde{W}(t)] \end{aligned}$$

Substitute $X(t) = e^{\int_0^t b(s)ds} \cdot R(t)$, we have

$$d(e^{\int_0^t b(s)ds} \cdot R(t)) = a(t)e^{\int_0^t b(s)ds}dt + \sigma(t)e^{\int_0^t b(s)ds}d\widetilde{W}(t)$$

Integrate both sides from time t to time T,

$$\int_t^T d(e^{\int_0^u b(s)ds} \cdot R(u)) = \int_t^T a(u)e^{\int_0^u b(s)ds}du + \sigma(u)e^{\int_0^u b(s)ds}d\widetilde{W}(u)$$

Stochastic Differential Equation Examples

Then, we have

$$\begin{aligned} & e^{\int_0^T b(s)ds} R(T) \\ = \quad & e^{\int_0^t b(s)ds} R(t) + \int_t^T a(u) e^{\int_0^u b(s)ds} du + \int_t^T \sigma(u) e^{\int_0^u b(s)ds} d\widetilde{W}(u) \end{aligned}$$

According to the initial condition $R(t) = r$, rearranging the terms, we get the solution of the SDE in the example 3;

$$\begin{aligned} & R(T) \\ = \quad & r e^{-\int_t^T b(s)ds} + \int_t^T a(u) e^{-\int_u^T b(s)ds} du + \int_t^T \sigma(u) e^{-\int_u^T b(s)ds} d\widetilde{W}(u) \end{aligned}$$

Here, we used the formulas $\int_0^t b(s)ds - \int_0^T b(s)ds = -\int_t^T b(s)ds$, $\int_0^u b(s)ds - \int_0^T b(s)ds = -\int_u^T b(s)ds$.

Stochastic Differential Equation Examples

Next, I will give the remarks of Example 3, talk about the distribution of $R(T)$ in the Hull-White model and its Markov Property.

Notes1: Now we have $R(T) = re^{-\int_t^T b(s)ds} + \int_t^T a(u)e^{-\int_u^T b(s)ds}du + \int_t^T \sigma(u)e^{-\int_u^T b(s)ds}d\widetilde{W}(u)$, calculate the expectation of $R(T)$, we have

$$\begin{aligned} E[R(T)] &= E[re^{-\int_t^T b(s)ds}] + E[\int_t^T a(u)e^{-\int_u^T b(s)ds}du] \\ &\quad + E[\int_t^T \sigma(u)e^{-\int_u^T b(s)ds}d\widetilde{W}(u)] \\ &= re^{-\int_t^T b(s)ds} + \int_t^T a(u)e^{-\int_u^T b(s)ds}du \\ &\quad + E[\text{Itô Integral}] \\ &= re^{-\int_t^T b(s)ds} + \int_t^T a(u)e^{-\int_u^T b(s)ds}du \end{aligned},$$

the last equation holds because the expected value of an Itô Integral is equal to 0.

Stochastic Differential Equation Examples

Notes2: Recall the property of an stochastic integral(or Itô integral), the details is written in the Blog 2. In the Hull-White model, the interest rate process $R(T)$ contains an Itô integral $\int_t^T \sigma(u)e^{-\int_u^T b(s)ds}d\widetilde{W}(u)$; fix time t, after change of variables, it also equals $\int_0^{T-t} \sigma(u-t)e^{-\int_{u-t}^T b(s)ds}d\widetilde{W}(u-t) = \int_0^{T-t} \sigma(v)e^{-\int_v^T b(s)ds}d\widetilde{W}(v)$

If we assume an integral condition $E[\int_0^\infty [\sigma(v)e^{-\int_v^T b(s)ds}]^2 dv] < \infty$ holds, then, the Itô integral $\int_0^{T-t} \sigma(v)e^{-\int_v^T b(s)ds}d\widetilde{W}(v)$ is a continuous martingale.

It holds that $E_S[\int_0^{T-t} \sigma(v)e^{-\int_v^T b(s)ds}d\widetilde{W}(v)] = \int_0^{S-t} \sigma(v)e^{-\int_v^T b(s)ds}d\widetilde{W}(v)$, for any time $S < T$.

Stochastic Differential Equation Examples

Notes3: At time t, $E[\int_t^T \sigma(u)e^{-\int_u^T b(s)ds} d\widetilde{W}(u)] = E[\int_t^T \sigma(u)e^{-\int_u^T b(s)ds} d\widetilde{W}(u)|\mathcal{F}_t]$ (because independent of the filtration $\mathcal{F}_t$)$= \int_t^t \sigma(u)e^{-\int_u^T b(s)ds} d\widetilde{W}(u) = 0$

Notes4: At time t, the variance of $R(T)$ is
$Var(R(T)) = Var(\text{non-random term} + \text{Itô integral}) = Var(\text{Itô integral}) = \int_t^T \sigma(u)^2 e^{-2\int_u^T b(s)ds} du$

It holds that $E_S[\int_0^{T-t} \sigma(v)e^{-\int_v^T b(s)ds} d\widetilde{W}(v)] = \int_0^{S-t} \sigma(v)e^{-\int_v^T b(s)ds} d\widetilde{W}(v) = 0$, for any time $S < T$. It follows from the variance formula and Itô isometry.

Notes5: It is noted that $R(T)$ is normally distributed with mean $E[R(T)]$ and $Var[R(T)]$. So, there is possibility $R(T)$ can be negative, and this is an issue with the Hull-White model because it allows negative interest rate.

Stochastic Differential Equation Examples

Notes6: Now Moving to discuss the Markov Property of the process $R(T)$. The solution process $R(T)$ is a Markov.

It means that if you have a function $h(R(T))$ and take the conditional expectation of $h(R(T))$ given information at time t, the conditional expectation only depends on the value of the process at time t; $R(t)$. It does not depend on the history of the process and only depend on the current value.

That is, it does not depend on which path the process has taken to reach $X(t)$, only depends on the value of the process at time t. Denote the dependency function by g, in mathematical notation, it is
$E_t[h(R(T))|\mathcal{F}_t] = g(R(t)) = g(t, R(t))$.

Fix the process initial value $R(t) = r$, then, we have
$E^{t,x}[h(X(T))] = E_t[h(X(T))|X(t) = x] = g(x) = g(t, x)$.

Stochastic Differential Equation Examples

Notes7: In summary, without a given initial condition, the stochastic process $R(T)$ is a **Markov**, because it only depends on the present value of $R(t)$ and are independent of the previous values of the process $\{R(s), s < t\}$.

With a given initial condition, the stochastic process $R(T)$ is independent of the previous values of the filtration $\{R(s), s \leq t\}$.